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Experiment Name: Handling missing and inappropriate data in a dataset.

```
import numpy as np
import pandas as pd

df = pd.read_csv("Hotel_Dataset.csv")
df
df.duplicated()
df.info()
df.drop_duplicates(inplace=True)
len(df)
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)

if 'Age_Group_1' in df.columns:
    df.drop('Age_Group_1', axis=1, inplace=True)
elif 'Age_Group.1' in df.columns:
    df.drop('Age_Group.1', axis=1, inplace=True)

df.loc[df['CustomerID'] < 0, 'CustomerID'] = np.nan
df.loc[df['Bill'] < 0, 'Bill'] = np.nan
df.loc[df['EstimatedSalary'] < 0, 'EstimatedSalary'] = np.nan
df.loc[
    (df['NoOfPax'] < 1) | (df['NoOfPax'] > 20),
    'NoOfPax'
] = np.nan
df.loc[
    (df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5),
    'Rating(1-5)'
] = np.nandf
['Age_Group'].unique()

df['Hotel'].unique()
df['Hotel'] = df['Hotel'].replace('Ibys', 'Ibis')
df['FoodPreference'].unique()
df['FoodPreference'] = df['FoodPreference'].replace(
    ['Vegetarian', 'veg', 'Veg'],
    'Veg'
)
df['FoodPreference'] = df['FoodPreference'].replace(
    ['non-Veg'],
    'Non-Veg'
)
df['EstimatedSalary'] = df['EstimatedSalary'].fillna(
    round(df['EstimatedSalary'].mean())
)
df['NoOfPax'] = df['NoOfPax'].fillna(
    round(df['NoOfPax'].median())
```

```
)
df['Rating(1-5)'] = df['Rating(1-5)'].fillna(
    round(df['Rating(1-5)'].median())
)
df['Bill'] = df['Bill'].fillna(
    round(df['Bill'].mean())
)
df
```

```
<class 'pandas.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     str
 2   Rating(1-5)      11 non-null     int64
 3   Hotel            11 non-null     str
 4   FoodPreference   11 non-null     str
 5   Bill             11 non-null     int64
 6   NoOfPax          11 non-null     int64
 7   EstimatedSalary  11 non-null     int64
 8   Age_Group_1      11 non-null     str
dtypes: int64(5), str(4)
memory usage: 924.0 bytes
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	6755.0	4.0	87777.0